

# Windows Write-Blocking Guide (CMD & Registry)

*Complete manual to write-protect USB and storage devices without external hardware*

---

There are two primary methods to **Write-Block (Read-Only)** any USB Drive, External Hard Disk, or Pen Drive in Windows using native tools. This guide provides a step-by-step walkthrough for both methods. Always open **Command Prompt (CMD)** as **Run as Administrator** before executing these commands.

## Method 1: Using Diskpart Utility (For a Specific USB Drive)

This method is highly recommended if you want to write-protect a specific pen drive currently connected to the computer while keeping other drives normal.

1. **Step 1:** Search for cmd in the Windows start menu, right-click on it, and select **Run as Administrator**.
2. **Step 2:** Launch the core disk utility by executing the following command:

```
diskpart
```

3. **Step 3:** List all the storage disks connected to the system:

```
list disk
```

*(Identify your USB drive's Disk Number, such as Disk 1 or Disk 2, by looking at its storage capacity/size.)*

4. **Step 4:** Select your target USB disk (Assuming your USB is Disk 1):

```
select disk 1
```

5. **Step 5:** Enable Write-Blocking (Read-Only mode) on the selected disk:

```
attributes disk set readonly
```

6. **Step 6:** Type exit and press Enter to leave the diskpart utility once completed.

### How to Revert Back to Normal?

If you need to write data or copy files to that USB drive again, follow the exact same steps above and replace Step 5 with the following command:

```
attributes disk clear readonly
```

## Method 2: Using Registry Editor (Global Policy for All USB Drives)

If you want **any USB drive connected to this computer** to automatically become write-blocked, you can apply a global Windows Registry policy. Since the required directory does not exist by default, the command below will automatically create it for you.

### 1. Command to Turn ON Write-Blocking:

Copy and paste this entire line into your Administrator CMD. This command automatically creates the folder structure and enables write-protection instantly:

```
reg add "HKLM\SYSTEM\CurrentControlSet\Control\StorageDevicePolicies" /f && reg add "HKLM\SYSTEM\CurrentControlSet\Control\StorageDevicePolicies" /v WriteProtect /t REG_DWORD /d 1 /f
```

*(After running the command, safely disconnect and re-insert your pen drive to see the write-protection in action.)*

### 2. Command to Turn OFF Write-Blocking:

To restore the system back to normal and allow standard data write access to all USB drives, run this command:

```
reg add "HKLM\SYSTEM\CurrentControlSet\Control\StorageDevicePolicies" /v WriteProtect /t REG_DWORD /d 0 /f
```

### Registry Editor (Manual Method)

If you prefer configuring this manually via the graphical Registry Editor (regedit):

- Navigate to: `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control`
- Right-click on **Control**, select **New -> Key**, and name it **StorageDevicePolicies**.
- Inside the new key, right-click the empty space, select **New -> DWORD (32-bit) Value**, and name it **WriteProtect**.
- Double-click **WriteProtect** and set its Value data to **1** (ON) or **0** (OFF).